TracexClean

Post-Exploitation Footprint Analysis and Cleanup Simulation Tool for Linux

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Abstract

Post-exploitation activities often leave behind forensic footprints such as authentication logs, shell histories, SSH artifacts, and temporary files. These traces play a crucial role in intrusion detection and incident response. However, beginners in cybersecurity tend to focus primarily on exploitation while neglecting post-exploitation awareness.

TracexClean is a Linux-based post-exploitation footprint analysis tool developed for educational and controlled lab environments. The tool identifies common forensic artifacts generated after attacker activities, evaluates their detection risk using a weighted scoring system, and provides ethical, non-destructive cleanup simulations. The project emphasizes security awareness, ethical responsibility, and defensive understanding of attacker behavior.

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1. Introduction

Post-exploitation refers to the phase after initial system compromise, where attackers perform actions such as privilege escalation, persistence, lateral movement, and data access. During this phase, multiple system artifacts and logs are generated that can be analyzed by defenders.

Understanding these artifacts is critical for:

- Detecting unauthorized access
- Performing forensic investigations
- Improving operational security awareness

TracexClean aims to bridge the knowledge gap by helping learners understand what footprints are created during post-exploitation and how they are detected.

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2. Problem Statement

Most beginner-level security tools and projects focus on exploitation techniques such as scanning, brute-forcing, or vulnerability exploitation. There is a lack of beginner-friendly tools that:

- Explain post-exploitation footprints
- Highlight forensic evidence generated by attacker actions
- Demonstrate ethical handling of sensitive system logs

This results in limited understanding of detection mechanisms and forensic analysis among learners.

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3. Objectives

The primary objectives of TracexClean are:

1. To identify post-exploitation footprints on Linux systems
2. To analyze forensic artifacts such as logs, histories, and SSH data
3. To evaluate detection risk using a scoring mechanism
4. To simulate cleanup actions without modifying system state
5. To promote ethical and responsible cybersecurity practices

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4. System Architecture

TracexClean follows a modular architecture to ensure clarity, maintainability, and extensibility.

Architecture Components

- **tracexclean.py**
  Acts as the main entry point, handling command-line arguments and execution flow.
- **scanner module**
  Performs read-only analysis of system logs and artifacts.
- **scorer module**
  Assigns weighted risk scores based on footprint severity.
- **reporter module**
  Generates human-readable terminal output and Markdown reports.
- **utility module**
  Handles permission validation and helper functions.

Design Philosophy

- Read-only analysis
- No destructive actions
- Ethical cleanup simulation
- Educational focus

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5. Implementation

5.1 Log Analysis

TracexClean analyzes Linux authentication logs (`/var/log/auth.log`) to detect:

- SSH login attempts
- Failed authentication attempts
- Sudo usage indicating privilege escalation

5.2 Artifact Detection

The tool detects common post-exploitation artifacts including:

- Bash command history files
- SSH artifacts such as `authorized_keys` and `known_hosts`
- Temporary files located in `/tmp` and `/var/tmp`

5.3 Risk Scoring System

Each detected footprint is assigned a predefined weight based on its forensic impact. The total score determines the overall risk level:

- Low Risk
- Medium Risk
- High Risk

This scoring system reflects operational noise rather than absolute compromise.

5.4 Cleanup Simulation

Instead of deleting logs or artifacts, TracexClean provides:

- Context-aware cleanup suggestions
- Ethical warnings
- Manual remediation guidance

No system files are modified during execution.

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6. Results and Analysis

The tool was tested on a personal Kali Linux system.

Observed Results

- SSH artifacts such as `known_hosts` were detected
- Multiple temporary files were identified
- Overall risk level was categorized as **Medium**

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[+] Starting TracexClean analysis ...

═══════════════════════════════════════════════════════════
 TracexClean - Post-Exploitation Footprint Report
═══════════════════════════════════════════════════════════
 Scan Time   : 2026-01-08 04:45:22.137977
 Risk Level  : Medium
 Total Score : 18
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[+] Detected Footprints:

1. Type      : SSH Artifact
   Severity  : High
   Log File  : /home/kali/.ssh/known_hosts
   Evidence  : known_hosts present in /home/kali/.ssh ...
 ──────────────────────────────────────────────────────────
2. Type      : Temp File
   Severity  : Medium
   Log File  : /tmp
   Evidence  : File found: /tmp/.xfsm-ICE-Z7O9H3 ...
 ──────────────────────────────────────────────────────────
3. Type      : Temp File
   Severity  : Medium
   Log File  : /tmp
   Evidence  : File found: /tmp/config-err-qAPZx3 ...
 ──────────────────────────────────────────────────────────
4. Type      : Temp File
   Severity  : Medium
   Log File  : /tmp
   Evidence  : File found: /tmp/.X0-lock ...
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[!] Cleanup Simulation (NO changes applied)

[!] Ethical Notice:
    Cleanup actions are simulated and should only be
    performed in controlled lab environments.

 - Audit ~/.ssh/authorized_keys for unauthorized persistence.
 - Inspect files in /tmp and /var/tmp. Unexpected binaries may indicate post-exploitation tools.

[+] Analysis complete.
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This outcome reflects realistic system behavior, demonstrating that not all detected artifacts indicate malicious activity.

Analysis

The results validate the effectiveness of TracexClean in distinguishing between normal system noise and potential post-exploitation indicators.

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7. Ethical Considerations

Ethical responsibility was a core focus during development. TracexClean:

- Does not delete or alter logs
- Does not clear user histories
- Does not automate cleanup actions

- Operates only in read-only or simulated modes

The tool is intended strictly for educational and controlled lab environments.

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8. Limitations

- Supports only Linux systems
- No real-time monitoring capabilities
- Cleanup actions are simulated, not enforced
- May flag legitimate system artifacts

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9. Future Enhancements

Future improvements may include:

- Windows platform support
- Rule-based detection using YAML
- Real-time monitoring mode
- Improved filtering of system-generated artifacts
- Integration with SIEM platforms

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10. Conclusion

TracexClean successfully demonstrates post-exploitation footprint awareness while maintaining ethical integrity. The project helped in understanding forensic artifacts, detection mechanisms, and responsible security practices. It serves as an effective educational tool for cybersecurity learners and reinforces the importance of post-exploitation analysis.

References

1. Linux Authentication Logs Documentation
2. MITRE ATT&CK Framework
3. OSCP Offensive Security Methodology
4. ChatGpt